CONFIDENTIAL

UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF COMPUTING

TEST 2 (PART 1)

SEMESTER I 2020/2021

SUBJECT CODE : SECJ/SCSJ2013
SUBJECT NAME : DATA STRUCTURES AND ALGORITHMS
YEAR/COURSE : 2 (SCSJ/ SCSV/ SCSB/ SCSR/ SCSP)
TIME : 20.10 – 21.20 MYT (1 hour 10 minutes)
DATE : 12th JANUARY 2021 (TUESDAY)

INSTRUCTIONS TO THE STUDENTS:

- Read the problem and instructions carefully.
- You are given **ONE HOUR TEN MINUTES** to complete the test inclusive of the submission of your program (**1 hour to answer** the question, **10 minutes to submit** the answer).
- Your program must follow the input and output as required in the text and shown in the examples. You must test the programs with (but not limited to) all the input given in the examples.

IMPORTANT NOTES:

- All the **COMMENT STATEMENTS** in the submitted program **WILL NOT BE EVALUATED**.

SUBMISSION PROCEDURE:

- Only the source code is required for the submission and the source code's file shall be named as follows: ***Name_ matricsNo_section.cpp*** (i.e. *AinaAli_A20EC018_01.cpp*).
- You do not need to compress the file.
- Submit the source code file via the **UTM's e-learning system**.

Question [40 Marks]

Given a class diagram for **`List`** as in **Figure 1**. Complete a linked list program given in **Program 1**. Define **`List`** class and write the implementation for all the function members for **`List`** to accomplish each of the following tasks:

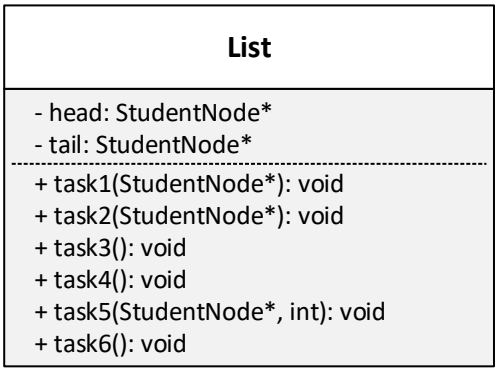


Figure 1: `List` class

```
1   //Program 1
2   class StudentNode {
3      public:
4         string name;
5         float CPA;
6         StudentNode *next, *prev;
7         StudentNode(string n=NULL, float c=0) {
8           name = n;
9           CPA = c;
10          next = NULL;
11          prev = NULL;
12        }
13  };
14
15  int main()
16  {
17    List StudList;
18    StudentNode *s1 = new StudentNode("Lisa", 3.56);
19    StudentNode *s2 = new StudentNode("Shah", 3.75);
20    StudentNode *s3 = new StudentNode("Amin", 3.67);
21    StudentNode *s4 = new StudentNode("Iman", 3.85);
22    StudentNode *s5 = new StudentNode("Afiq", 3.61);
23
24    cout << fixed << setprecision(2) << left;
25    StudList.task1(s1);
26    StudList.task2(s2);
27    StudList.task2(s3);
28    StudList.task3();
29    StudList.task4();
30    StudList.task3();
```

```
31      StudList.task5(s4, 2);
32      StudList.task5(s5, 3);
33      StudList.task3();
34      StudList.task6();
35      StudList.task3();
36
37      return 0;
38  }
```

Task 1: Define a function named **`task1`**. The function should able to insert a new student node into an empty list. **Figure 2** shows the result of this task after the execution of the statement in **Line 25**. **(5 marks)**

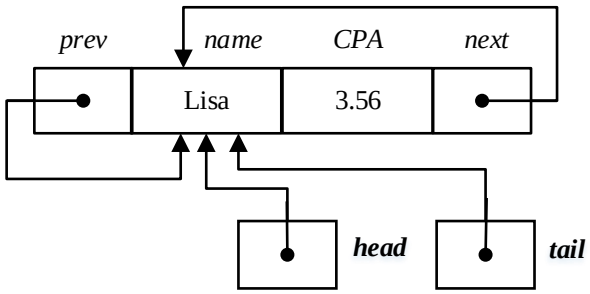


Figure 2: Current node in the list after insertion node in the empty list

Task 2: Define a function named **`task2`**. The function should able to insert a new student node at the end of the list. **Figure 3** shows the result of this task after the execution of the statement in **Line 26** and **Figure 4** shows the result of this task after the execution of the statement in **Line 27**. **(5 marks)**

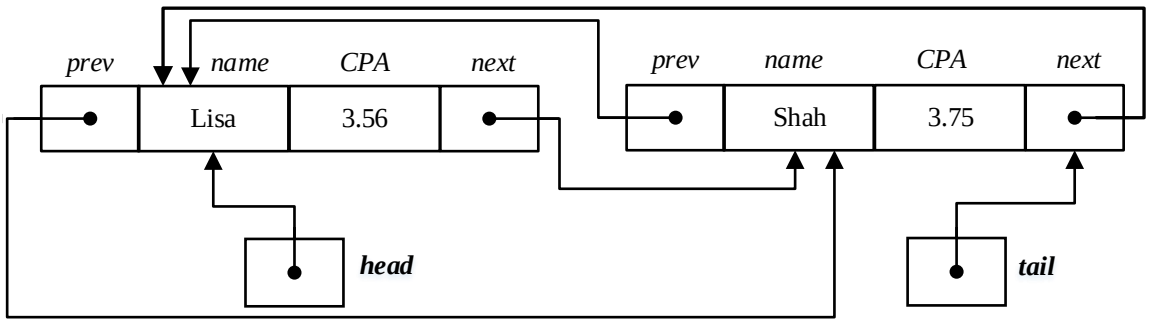


Figure 3: Current nodes in the list after the first insertion node at the end of the list

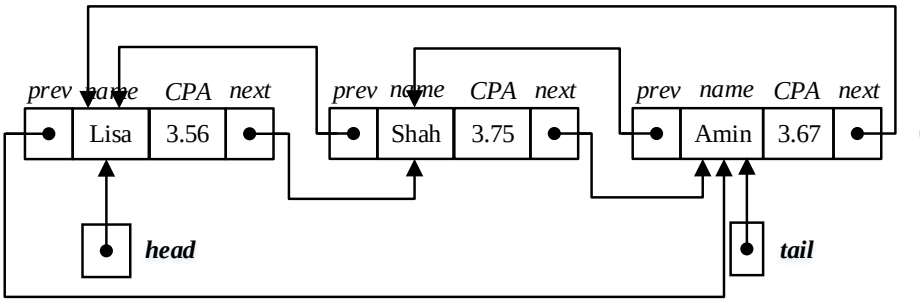


Figure 4: Current nodes in the list after the second insertion node at the end of the list

Task 3: Define a function named **`task3`**. The function should able to print all data in the list in forward order. **Figure 5** shows the result of this task after the execution of the statement in **Line 28**. ***Note:*** In this task, you should use a **loop**. **(7 marks)**

```
Print forward:
[Lisa    3.56] -> [Shah    3.75] -> [Amin    3.67]
```

Figure 5: Output after the execution of the statement in Line 28

Task 4: Define a function named **`task4`**. The function should able to delete the first node in the list. **Figure 6** shows the result of this task after the execution of the statement in **Line 29**. **(6 marks)**

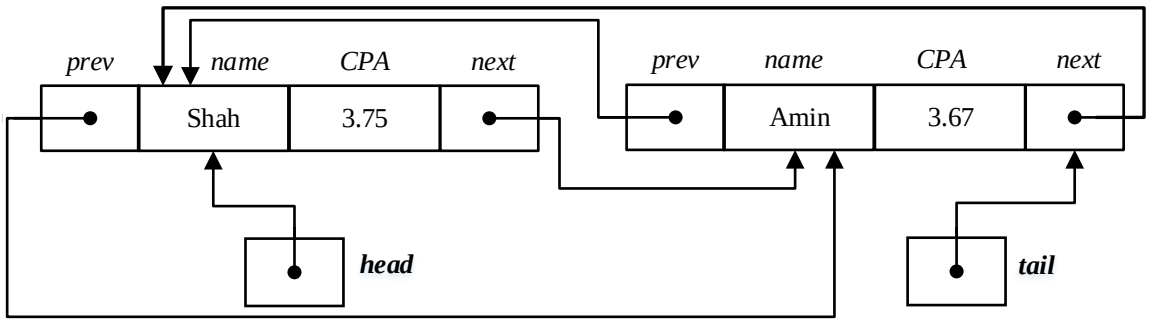


Figure 6: Current nodes in the list after the deletion of the first node in the list

Task 5: Define a function named **`task5`**. The function should able to insert a new student node at the middle of the list. The position of the new node in the list depends on the **value of the second argument** of this function. Assume the value of the second argument of this function is 2. This means that the new node will be the second node in the list (being in

the second position in the list). **Figure 7** shows the result of this task after the execution of the statement in **Line 31** and **Figure 8** shows the result of this task after the execution of the statement in **Line 32.** ***Note:*** In this task, you should use a **loop**. **(11 marks)**

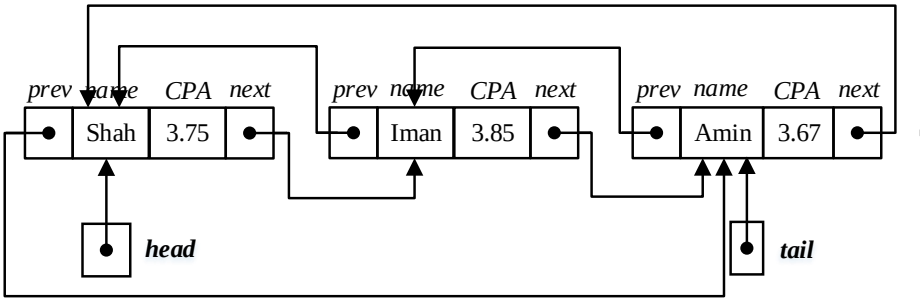


Figure 7: Current nodes in the list after the insertion node at the second position in the list

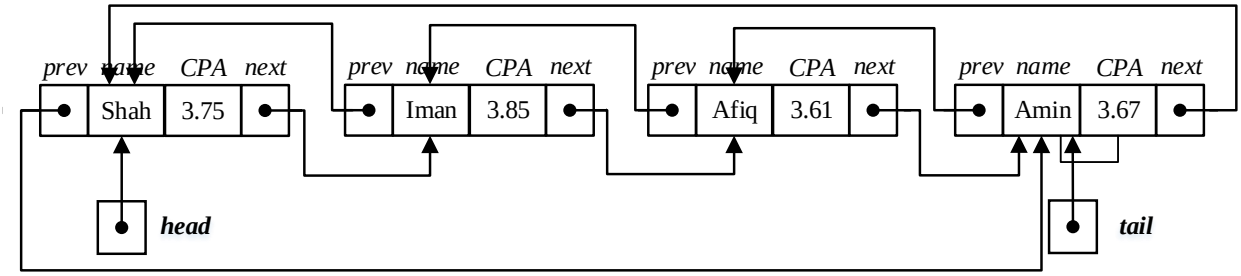


Figure 8: Current nodes in the list after the insertion node at the third position in the list

Task 6: Define a function named **`task6`**. The function should able to delete the last node in the list. **Figure 9** shows the result of this task after the execution of the statement in **Line 34**. **(6 marks)**

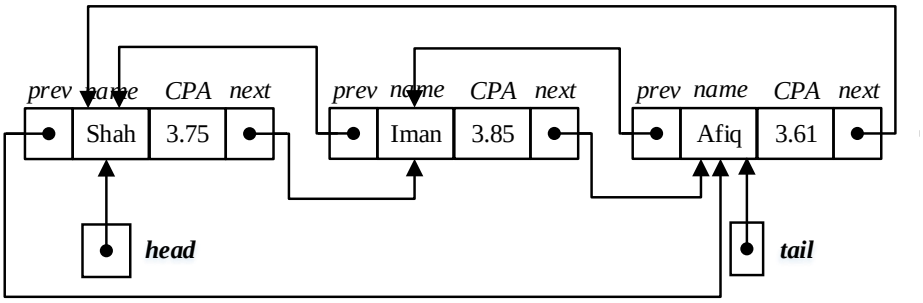


Figure 9: Current nodes in the list after the deletion of the last node in the list

After completing all tasks, please make sure your program should be able to produce the output as shown in **Figure 10.**

```
Print forward:
[Lisa    3.56] -> [Shah    3.75] -> [Amin    3.67]

Print forward:
[Shah    3.75] -> [Amin    3.67]

Print forward:
[Shah    3.75] -> [Iman    3.85] -> [Afiq    3.61] -> [Amin    3.67]

Print forward:
[Shah    3.75] -> [Iman    3.85] -> [Afiq    3.61]
```

Figure 8: The output of completed Program 1